Passenger/Crew – General Data

1. General

Two crew entry doors, one each side of the flight compartment, and two passenger doors, one each side of the cabin compartment are provided (refer to Figure 2).

A. Flight Compartment Doors

Each flight compartment door has a fixed window and a sliding window with a release handle. The doors are secured to the airframe structure by two hinges at the forward edge of each door, which allows either door to be opened outward and forward, to a position limited by a nylon restraining strap. An interior and exterior door handle is installed, which when turned clockwise, operates a latch mechanism which disengages a mid-door bolt, and an upper latch pin. Each exterior door handle can be locked by key from the outside. A map case is provided in the lower part of each door, and when a crew oxygen system is installed, clips are installed at the center of the door to support the flexible oxygen hoses.

B. Cabin Compartment Doors

Refer to Figure 1 for details.

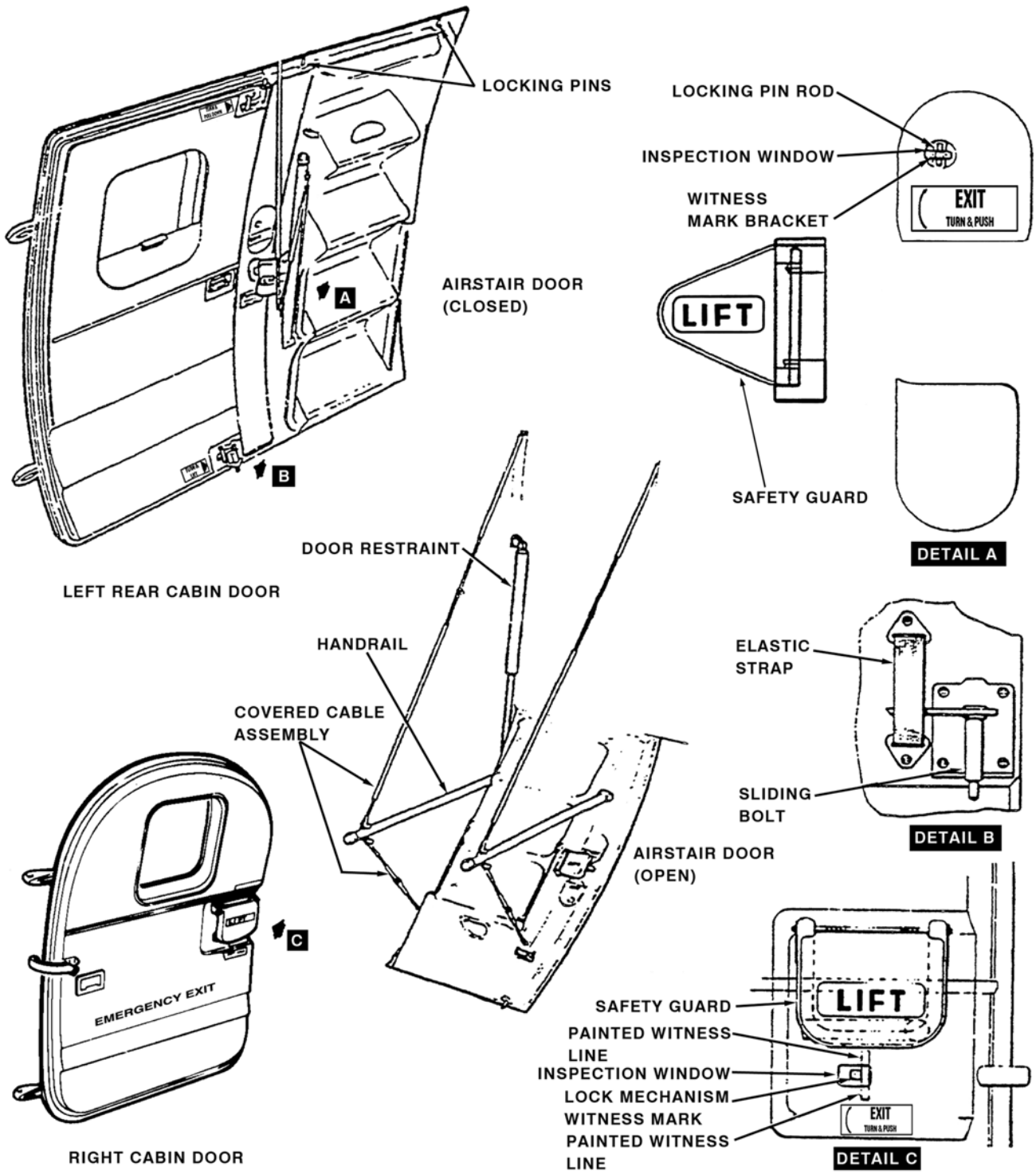
The cabin left compartment door is in two parts; the forward part is an airstair door for passenger entry, and the rear part to provide a larger entrance space for cargo loading.

The airstair door as installed to the basic aircraft, opens downward and outward, and is supported in the open position by two covered cable assemblies. One cable assembly is connected to the front edge of the door, and the other to the rear; both cables run parallel with each other and attach to the fuselage structure at the upper door frame. Aircraft have an adjustable eye fitting with a locknut on the rear cable assembly. The fitting is threaded onto the upper end of the rear cable assembly and connected to the rear door frame. The fitting can be adjusted to equalize cable tension between the front and rear cable assemblies. The airstair door is provided with a door restraint (door saver) which prevents the door falling free when being lowered and gives spring assist when being raised. The airstair door can be quickly removed to provide easier access for cargo loading. To assist in passenger entry and exit, the airstair door is provided with handrails. Each handrail is attached between its associated covered cable assembly and the appropriate side of the airstair door, and is so designed that as the door is closed, it will fold up against the inner side of the door. The airstair door has an internal and external handle which operates a latch mechanism to engage the two door locking pins in the fuselage upper door surround structure. An inspection window provides the means of checking that the two fluorescent orange dayglo paint witness marks (one on the locking pin rod and the other on a fixed bracket), align when the door is fully locked (refer to Figure 1, Detail A). Contacts, to complete the electrical circuit to the internal door unlocked microswitch (refer to 52-70-00, Doors Unlocked Caution System – General Data), are located on both the door and the fuselage door upper surround.

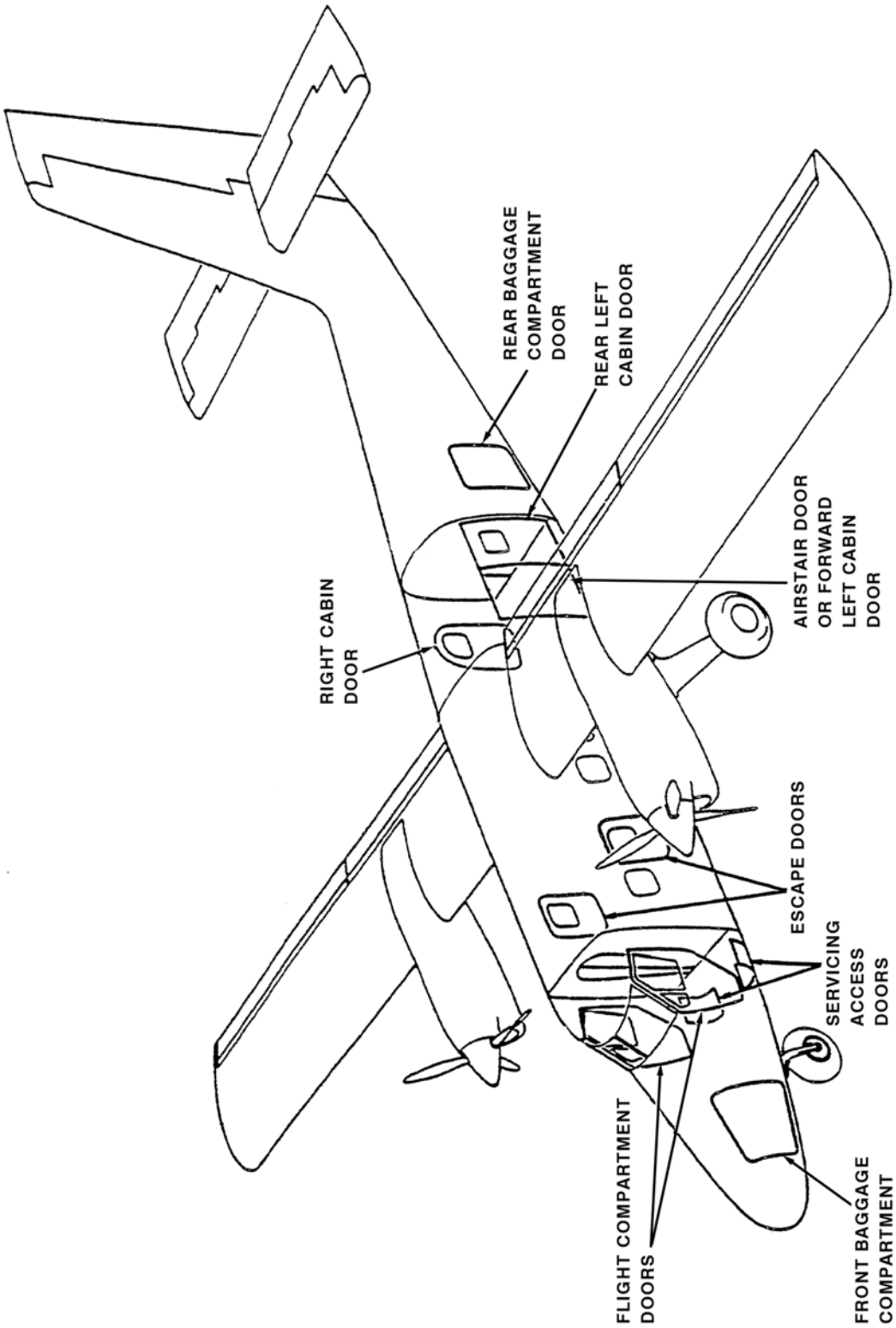
The cabin left rear door hinged to the rear part of the door frame, opens outward and rearward. When closed, the door is locked top and bottom by engaging sliding bolts which are then secured with elastic straps.

The cabin right door is hinged at the forward part of the door frame and incorporates a strap assembly which restricts door movement and prevents it from contacting and causing damage to the fuselage skin when the door is in the open position. The door latch mechanism, operated by an internal or external handle, has a latch bolt which engages in the door rear surround structure. An inspection window adjacent to the door lock handle shows, when the door is locked, a part of the door latch mechanism marked with fluorescent orange dayglo paint. With the door locked, this painted portion should be aligned with painted lines on each side of the inspection window.

The exterior cabin door handles can be locked by a key from the outside, and each interior handle is covered by a spring-loaded plastic safety guard which must be lifted in order to operate the handle.



Cabin Doors, Locks and Safety Guards
Figure 1



Doors – Location
Figure 2

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2. Removal/Installation

A. Remove Airstair Door (Mod 6/2024)

1) Open and support airstair door.

2) Remove hardware attaching door saver to airstair and separate door saver from airstair.

3) Remove hardware attaching airstair handrail cables from aircraft fuselage.

4) Remove screws attaching piano hinge retainer and DZUS or camlock fasteners from aircraft and remove door.

5) Remove inner handle guard assembly (Mod 6/2326).

a) Remove screw attaching inner handle guard assembly and retain hardware.

b) Disassemble guard assembly by removing cotter pin at both ends of rod and retain.

c) Remove and discard damage springs.

d) Install new springs and assembly guard assembly by securing rod both ends with cotter pin.

B. Install Airstair Door (Mod 6/2024)

1) Position and support airstair door and align mounting holes of piano hinge to fuselage.

2) Secure piano hinge to fuselage with screws as required.

3) Secure airstair handrail cables to fuselage with attaching hardware.

4) Secure door saver to airstair door with attaching hardware.

5) Verify electrical conductivity between door and airframe. Refer to Para 3.C.

6) Install inner handle guard assembly (Mod 6/2326).

a) Install inner handle guard assembly, using retained hardware (refer to Para 2.A.5)).

b) Close airstair door and verify inner handle guard assembly is in contact with the handle.

C. Remove Airstair Door (Mod 6/2295)

1) Open and support airstair door.

2) Remove quick release pin attaching airstair door saver to fuselage.

3) Remove quick release pin attaching handrail cables to the door handrail.

4) Remove screws attaching piano hinge retainer and DZUS or camlock fasteners from aircraft and remove door.

D. Install Airstair Door (Mod 6/2295)

1) Position and support airstair door and align mounting holes of piano hinge to fuselage.

2) Secure piano hinge to fuselage with screws as required.

3) Secure airstair handrail cables to fuselage with quick release pin.

4) Secure airstair door saver to fuselage with quick release pin.

5) Verify electrical conductivity between door and airframe. Refer to Para 3.C.

E. Remove Cabin Left Rear Door

1) Open and support Cabin Left Rear Door.

2) Remove hinge pins attaching door to airframe and remove the door.

F. Install Cabin Left Rear Door

1) Install hinge pins attaching door to airframe.

2) Close door and check fit and alignment to airframe.

3) Verify electrical bonding of the door to airframe is maintained. Refer to Para 3.E.

G. Remove Cabin Right Rear Door

1) Open and support Cabin Right Rear Door.

2) Remove hinge pins attaching door to airframe and remove the door.

H. Install Cabin Right Rear Door

1) Install hinge pins attaching door to airframe.

2) Close door and check fit and alignment to airframe.

3) Verify electrical bonding of the door to airframe is maintained. Refer to Para 3.E.

I. Remove Flight Compartment Door

1) Open and support Flight Compartment Door.

2) Remove hinge pins attaching door to airframe and remove the door.

J. Install Cabin Flight Compartment Door

1) Install hinge pins attaching door to airframe.

2) Close door and check fit and alignment to airframe.

3) Verify electrical bonding of the door to airframe is maintained. Refer to Para 3.G.

3. Adjustment

Note Whenever airstair door is serviced or removed from aircraft, adjustment/check of the airstair door latch pin engagement is required.

Note Tapered section of latch pins is to be ignored when determining engagement.

Note If screws attaching the piano hinge or latch pin striker plates to the door structure are removed they must be reinstalled wet using PR–1764 CLASS B electrically conductive sealant with PR–148 adhesion promoter or equivalent.

A. Adjust the amount of latch pin engagement of the airstair door by adjusting the latch pin push rod assembly.

B. Ensure pin engagement on latch pins when door is in the closed position is minimum 0.25 of an inch.

C. Airstair Door Bonding Test

With door closed, measure the resistance between the airstair door and the airframe using a milliohm meter, resistance to be less than 10 milliohms. Probe measurements to be taken on one of the airstair door hinge fasteners and an airframe side rivet. Refer to Figure 3.

D. Airstair Door Bonding Adjustment/Rework

Remove the Airstair door hinge attachment fasteners and clean as required to achieve bonding requirement. For reference, refer to applicable standard electrical bonding procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

E. Cabin Left/Right Rear Doors Bonding Test

With door closed, measure the resistance between the rear cabin doors and the airframe using a milliohm meter, resistance to be less than 10 milliohms. Probe measurements to be taken on one of the rear door hinge fasteners and an airframe side rivet. Refer to Figure 4.

F. Cabin Left/Right Rear Doors Adjustment/Rework

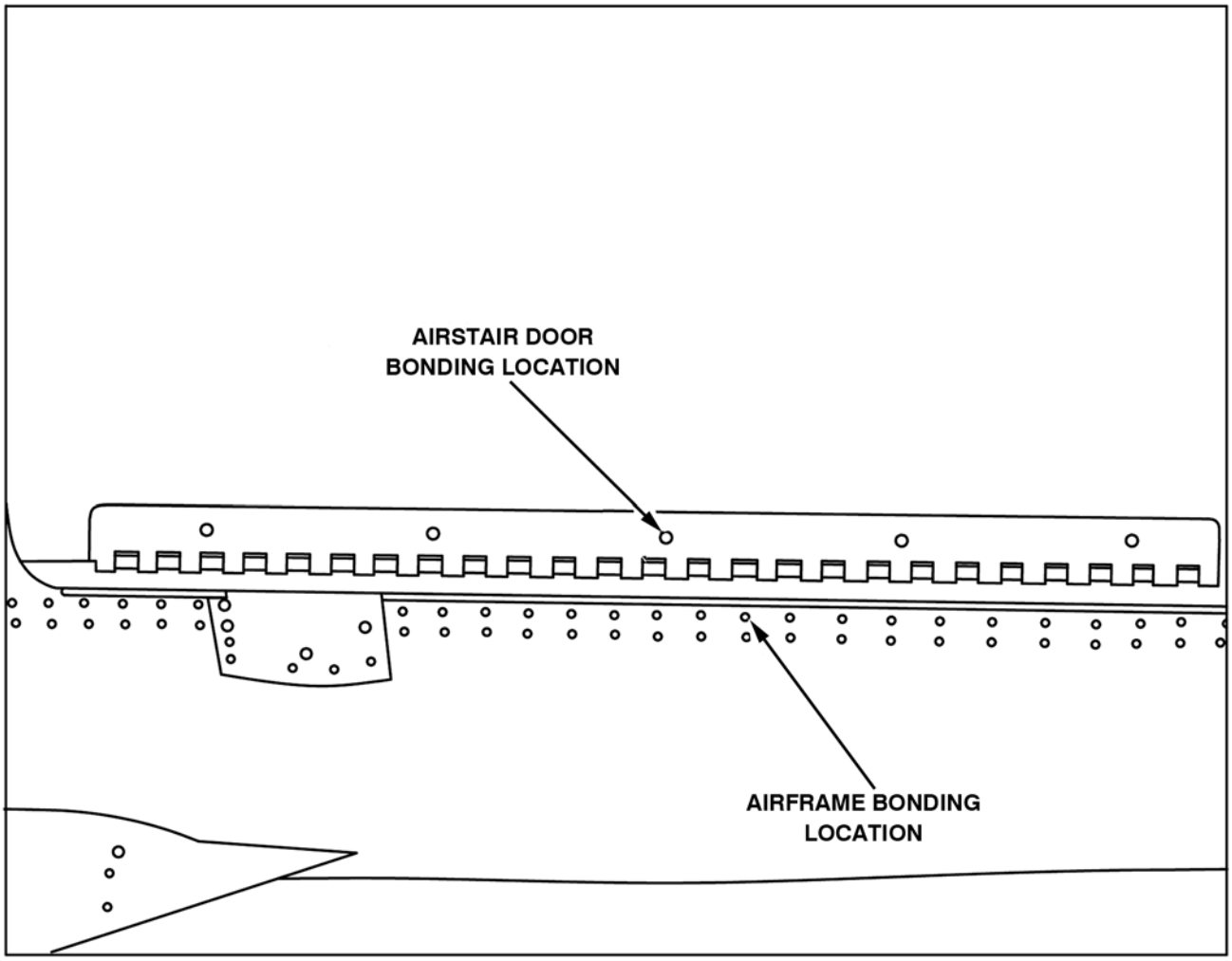
Remove the rear cabin door hinge attachment fasteners and clean as required to achieve bonding requirement. If required, remove rear cabin door to airframe hinge pin and clean to achieve bonding requirement. For reference, refer to applicable standard electrical bonding procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

G. Flight Compartment Doors Bonding Test

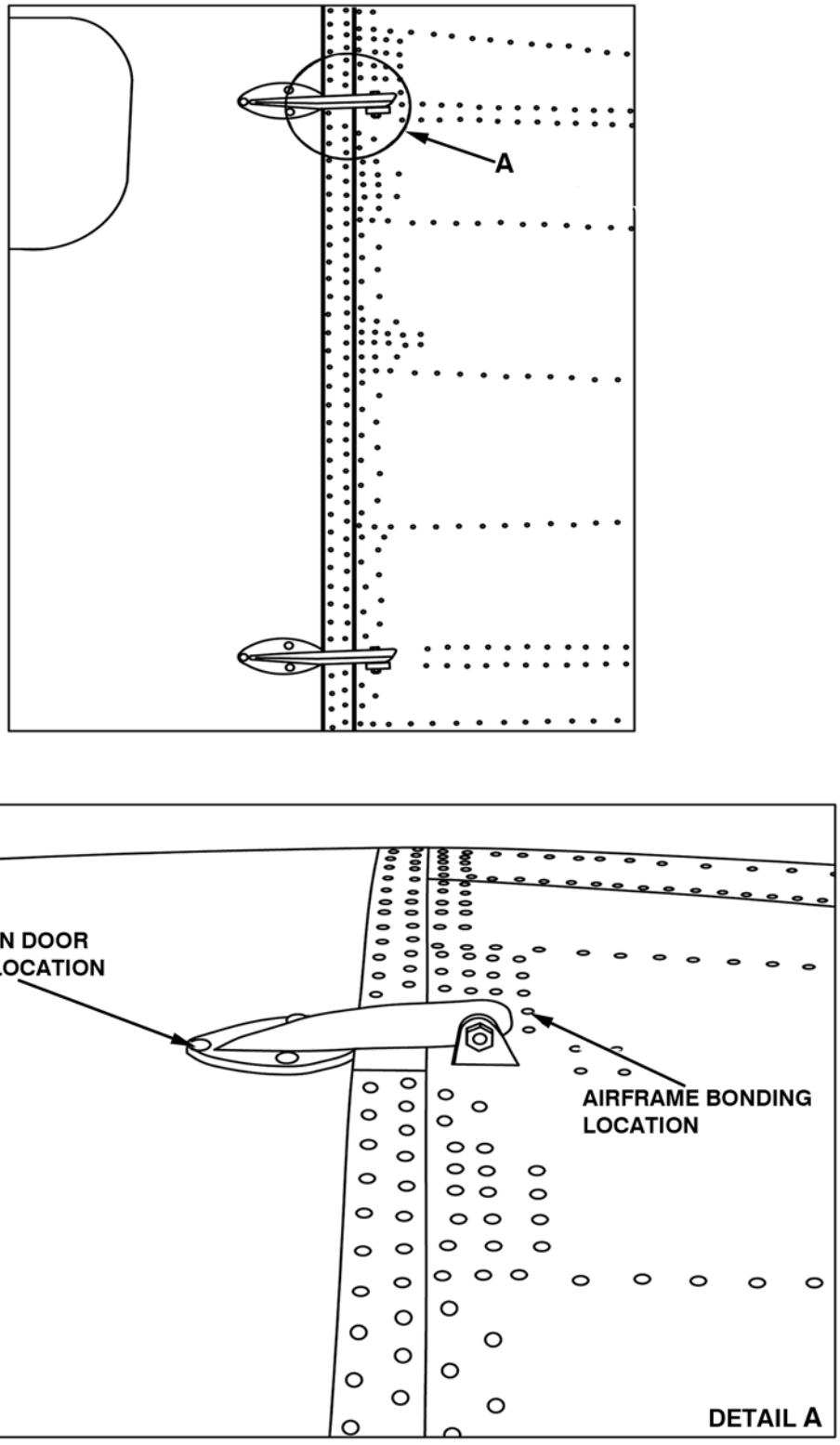
With door closed, measure the resistance between the flight compartment door and the airframe using a milliohm meter, resistance to be less than 10 milliohms. Probe measurements to be taken on one of the door hinge fasteners and an airframe side rivet. Refer to Figure 5.

H. Flight Compartment Doors Bonding Adjustment/Rework

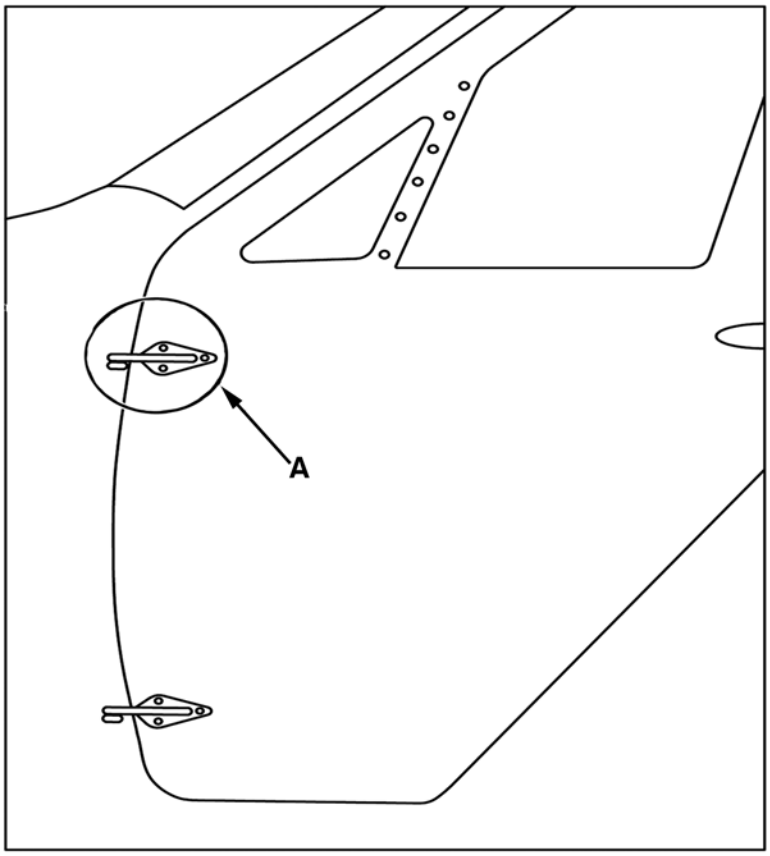
Remove the door hinge attachment fasteners and clean as required to achieve bonding requirement. If required, remove door to airframe hinge pin and clean to achieve bonding requirement. For reference, refer to applicable standard electrical bonding procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

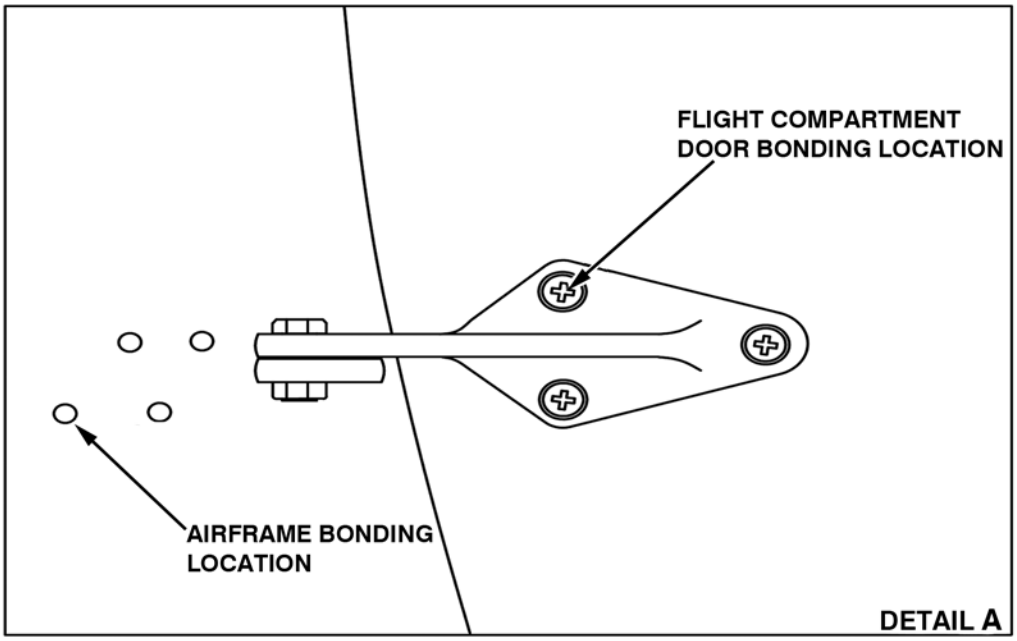


Airstair Door – Electrical Bonding Check Locations
Figure 3



Rear Cabin Doors – Electrical Bonding Check Locations (Left Door Shown)
Figure 4





Flight Compartment Doors – Electrical Bonding Check Locations (Left Door Shown)
Figure 5

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